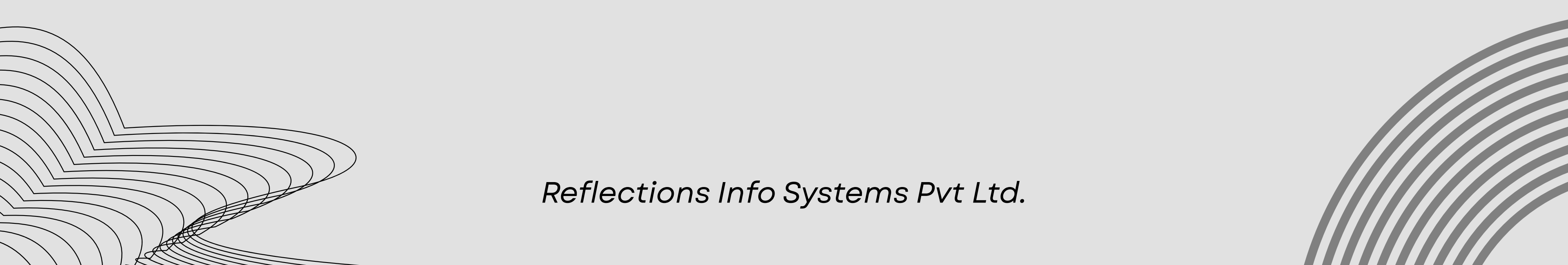




AI CHATBOT & RAG AI SYSTEM

by Janaki R



Reflections Info Systems Pvt Ltd.

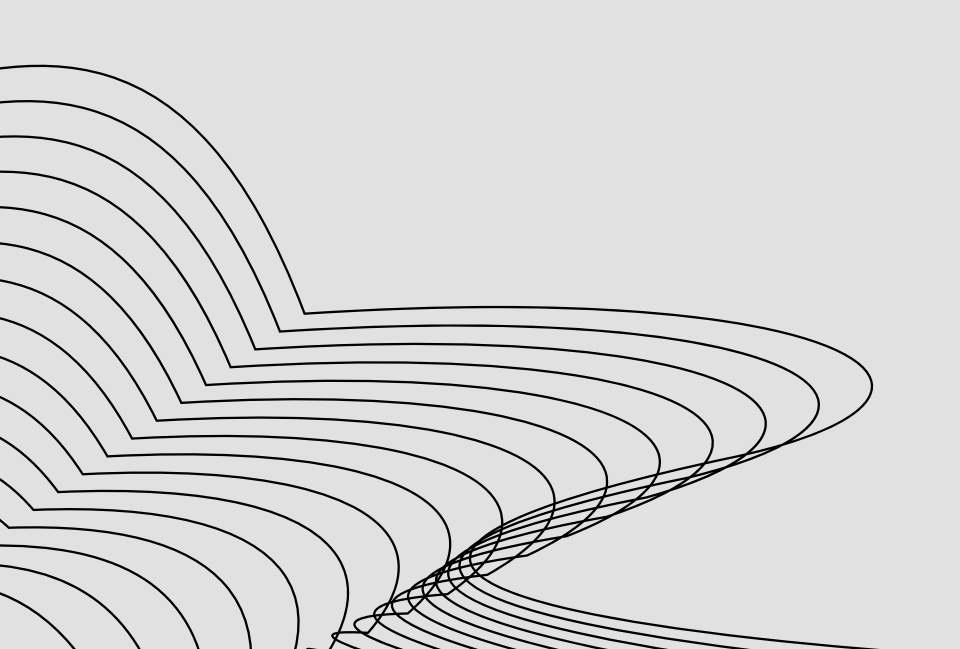
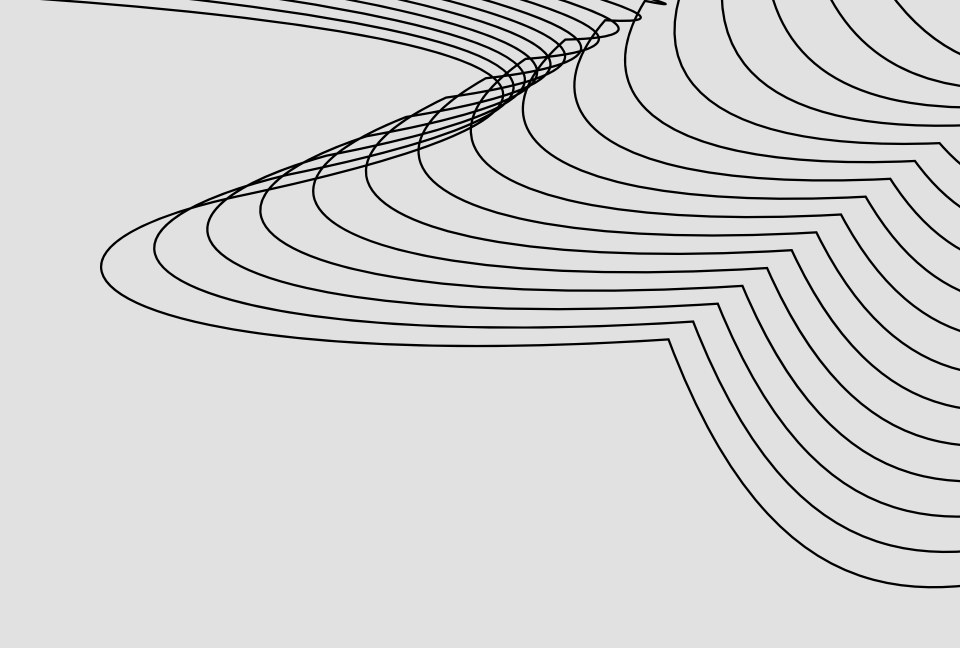


PROJECT OVERVIEW

This presentation covers two AI-based projects:

1. AI Chatbot using Groq API
2. RAG-based PDF Chatbot

These projects demonstrate how AI can improve automated communication and document-based information retrieval.



Task 1:

AI Chatbot

Task 1

Overview

The AI Chatbot is a simple conversational assistant built using the Groq API.

Main functions:

- Answer user questions
- Maintains conversation history
- Generates real-time responses

The project helped in understanding how Large Language Models (LLMs) work.

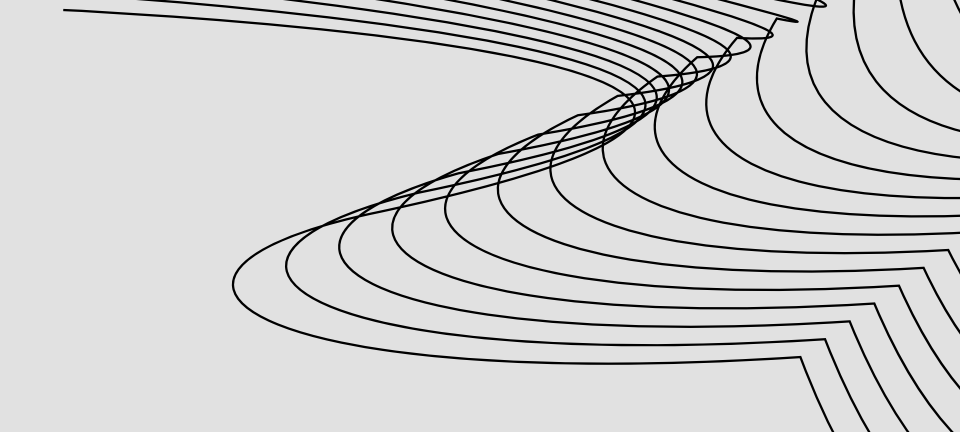
BUSINESS PURPOSE & FEATURES

Businesses often need automated systems to respond to user queries quickly.

- Businesses need faster customer interaction systems
- Manual responses take time
- Repetitive queries increase workload

Key Features:

- Real-time AI responses
- Multi-turn conversation memory
- Adjustable AI behaviour
- Secure API key handling using .env



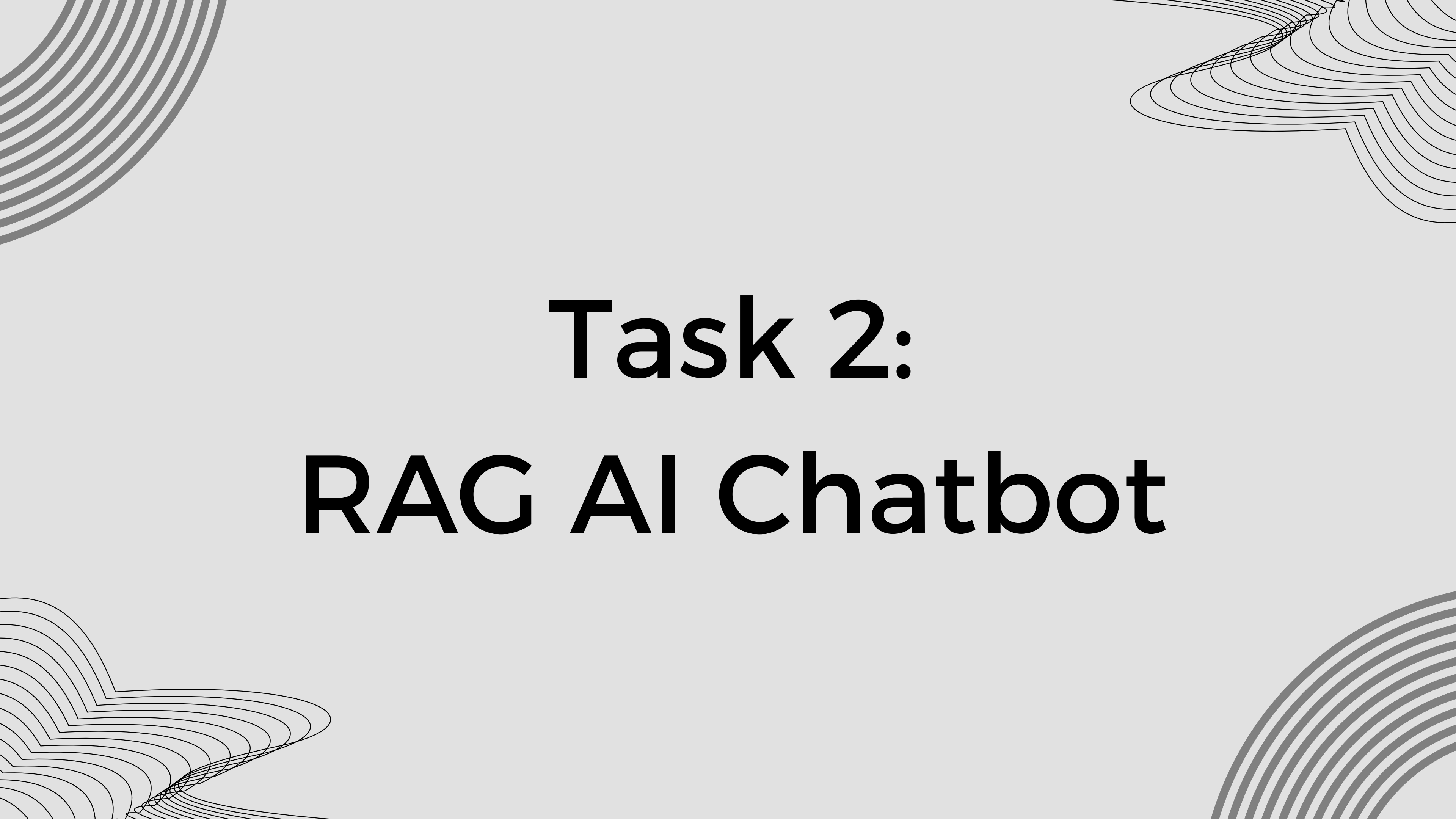
CONTRIBUTION & CHALLENGES

My Contribution:

- Integrated Groq API
- Implemented chat history memory
- Added prompt customization
- Managed API security

Challenges Faced:

- Managing conversation history(Solved by storing messages in a messages list)
- Deprecated Model Issue (Solved by replacing the deprecated model with llama-3.1-8b-instant)

The background features decorative wavy lines in the corners. The top-left and bottom-right corners have thick, dark gray wavy lines. The top-right and bottom-left corners have thin, light gray wavy lines.

Task 2:

RAG AI Chatbot

Task 2

Overview

The RAG AI System is a document-based AI chatbot that answers questions using uploaded PDF files.

Unlike a normal chatbot, this system retrieves relevant information from documents before generating responses.

Technologies used:

- LangChain
- ChromaDB
- HuggingFace Embeddings
- Groq API
- Streamlit

BUSINESS PURPOSE & FEATURES

Organizations store a large amount of information in PDFs and reports.

- Searching manually takes time
- Important information is difficult to locate quickly
- Large PDFs are hard to navigate

The RAG system allows users to ask questions directly from documents and receive accurate context-based answers instantly.

Key Features are:

- Multiple PDF support
- Semantic search
- Chat history memory
- Streamlit web interface
- Source chunk display
- Persistent local database
- Context-aware AI responses

HOW THE RAG SYSTEM WORKS

PDF Documents

Input documents for processing

Vector Database Storage

Storing embeddings for efficient retrieval

Semantic Search Retrieval

Finding relevant document sections

Text Chunking

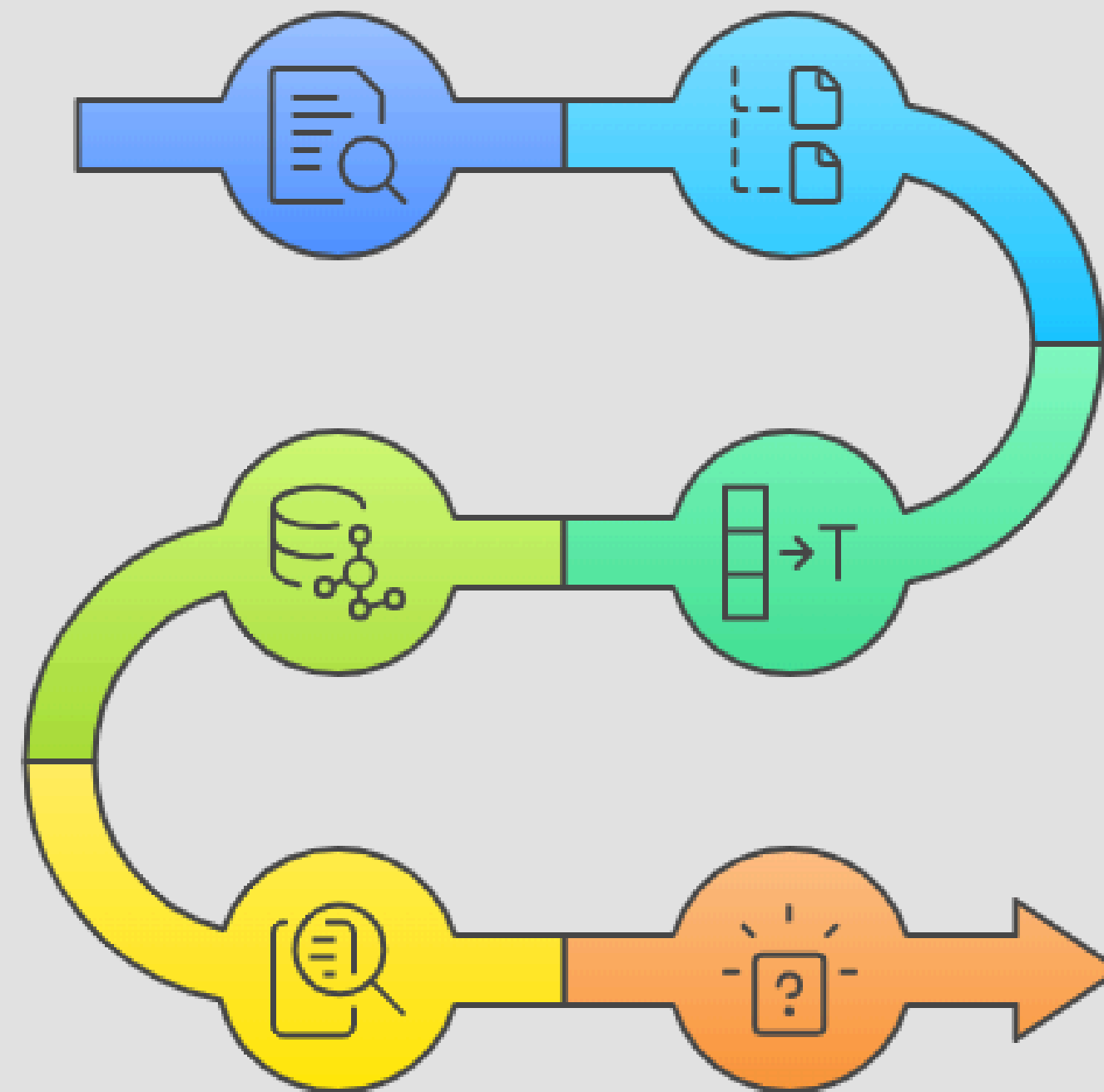
Dividing documents into smaller sections

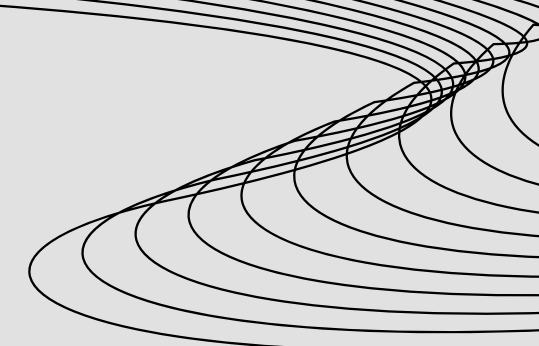
Embeddings Generation

Converting text chunks into numerical vectors

AI Generates Final Answer

Creating answers based on retrieved context





CONTRIBUTION & CHALLENGES

My Contribution:

- Built the complete RAG pipeline
- Implemented PDF chunking and embeddings
- Integrated ChromaDB vector database
- Developed the Streamlit frontend UI
- Added logging and persistent storage

Challenges Faced:

- Understanding Vector Similarity (Solved by learning how embeddings capture meaning)
- Slow embedding generation (Solved using persistent ChromaDB storage)
- Chat history reset during reloads (Solved by using session_state in Streamlit)

RESULTS

Testing Performed:

- Asked multiple PDF-related questions
- Verified correct chunk retrieval
- Compared answers with and without RAG
- Tested different retrieval settings

Observations:

- RAG produced more accurate answers
- Semantic search improved retrieval quality
- Streamlit improved usability

SCREENSHOTS

AI Chatbot



whats acid rain?



Acid rain is a significant global environmental problem that has a negative impact on various ecosystems. It is formed when water molecules in the atmosphere combine with SO_2 , typically near industrial areas. This type of rain is injurious to humans, plants, and certain types of rocks, and can corrode metals, damage buildings, and monuments. Acid rain also poses health risks by allowing sulphate particles to penetrate deep into the lungs.



explain photochemical smog



Photochemical smog, also known as Los Angeles smog, occurs in warm, dry, and sunny climates. It was first observed in the 1940s in Los Angeles. The main components of photochemical smog are formed when sunlight acts on unsaturated hydrocarbons and nitrogen oxides produced by automobiles and factories. This results in a high concentration of oxidizing agents, making it called as oxidizing smog. It can cause respiratory issues and eye discomfort.

> Retrieved Chunks

Ask a question



Ask a question: what is the difference between leadership and management?

Retrieved Chunks:

--- Chunk 1 ---

Leadership and management are not the same.

- Leadership = influencing people
- Management = planning, organizing, controlling

A person can:

- lead without formal authority
- manage without inspiring others

An individual can influence and guide others without holding any official position.

Example: College . A student who:

- motivates classmates
- organizes group study
- helps others understand concepts

Not officially appointed → but still a leader

Example: Company . A senior employee who:

--- Chunk 2 ---

Leadership and management are not the same.

- Leadership = influencing people
- Management = planning, organizing, controlling

--- Chunk 3 ---

Leadership and management are not the same.

- Leadership = influencing people
- Management = planning, organizing, controlling

A person can:

- lead without formal authority
- manage without inspiring others

An individual can influence and guide others without holding any official position.

Example: College . A student who:

- motivates classmates
- organizes group study
- helps others understand concepts

Not officially appointed → but still a leader

Example: Company . A senior employee who:

AI Answer:

The difference between leadership and management is that leadership is about influencing people, whereas management is about planning, organizing, and controlling. A person can lead without holding formal authority, and they can also manage without inspiring others. In other words, leadership is about guiding and motivating others, while management is about overseeing and directing tasks. This distinction is highlighted by the examples of a student who motivates classmates and organizes group study, and a senior employee who leads without an official position.

FUTURE IMPROVEMENTS

- Support more file formats
- Improve retrieval accuracy
- Add file upload support
- Improve the Streamlit UI
- Add source citations
- Optimize retrieval speed
- Add response feedback system for users

CONCLUSION

These projects helped in understanding:

- Conversational AI systems
- Retrieval-Augmented Generation (RAG)
- Semantic search and vector databases
- Context-aware AI applications

The projects also provided practical experience in building real-world AI applications using modern AI tools and frameworks.

The background is a light gray color. In the four corners, there are decorative elements consisting of multiple concentric, curved lines. The lines in the top-left and bottom-right corners are a medium gray, while the lines in the top-right and bottom-left corners are a lighter gray. These lines create a sense of depth and movement, framing the central text.

THANK YOU